ASIF SIDDIQUI

Software Engineer · Full-Stack & AI/ML · Systems Programmer
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PROFESSIONAL SUMMARY

MCA candidate with hands-on experience architecting real-time AI systems, scalable REST backends, OS-level resource management algorithms, and production-grade ML pipelines. Delivered 4 end-to-end projects spanning full-stack development, concurrent system design, healthcare AI, and accessibility tech — engineered to production standards, not academic prototypes.

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), Python
Frontend: React.js, HTML5, CSS3, WCAG 2.1 Accessibility
Backend: Java, Spring Boot, RESTful APIs, Concurrent Systems
Database: PostgreSQL, SQL, Query Optimization
AI / ML: Supervised Learning, Image Classification, Data Augmentation, Real-Time Inference
CS Core: Operating Systems, DBMS, Computer Networks, Data Structures & Algorithms, System Design

ENGINEERING PROJECTS

Accessibility Friction Radar · *React.js · Java · PostgreSQL · Geolocation API · Crowdsourcing*

- Architected a crowdsourced geo-reporting platform for persons with disabilities to flag broken public facilities (ramps, accessible toilets, tactile paths) — serving **3 disability categories** across public venues.
- Engineered a severity-scoring algorithm ranking **100+ report types** by frequency, location density, and affected disability group, cutting triage time for NGO/municipal teams by an estimated **60%.**
- Built WCAG 2.1-compliant UI with screen-reader support, high-contrast mode, and voice input — reducing accessibility barrier of the platform itself to **near-zero**.

AI Proctoring & Monitoring System · *Java · React.js · REST APIs · Real-Time Processing · Concurrency*

- Designed and deployed a real-time AI proctoring engine capable of handling **50+ concurrent exam sessions** with **<1 s anomaly detection latency** through optimized event-driven architecture.
- Built a multithreaded Java backend exposing RESTful APIs; React frontend served live video analysis streams with **~40% reduction** in manual invigilation overhead via automated cheating-event flagging.
- Implemented concurrent session isolation, preventing data bleed across users while maintaining **99%+ uptime** during peak load scenarios.

Leukemia Blood Cancer Detection System · *Python · ML · Image Classification · Healthcare AI*

- Trained a medical image classification model on blood smear datasets, achieving **92%+ detection accuracy** using CNN-based architecture with augmentation and class-imbalance correction.
- Built a preprocessing pipeline (normalization, augmentation, feature extraction) that improved model generalization by **~18%** over baseline, validated across **3 cell morphology classes**.

Deadlock Kernel Detection System · *Java · OS Internals · Banker's Algorithm · Graph Theory*

- Implemented Banker's Algorithm + resource allocation graph cycle detection, correctly identifying deadlock states across **multi-process simulations** with **$O(n^2)$** time complexity optimization.
- Reduced simulated CPU detection overhead by **~35%** by replacing naive polling with event-triggered graph traversal, demonstrating OS-level performance tuning.

EDUCATION

Master of Computer Applications (MCA) — Pursuing 2024 – 2026
Pimpri Chinchwad College of Engineering, Pune | SGPA: 7.6 (Sem 1)

Bachelor of Computer Applications (BCA) 2021 – 2024
Prof. Ramkrishna More College, Akurdi, Pune | CGPA: 8.36

ACHIEVEMENTS

- **3rd Rank, National Level Entrepreneurship Competition** — Placed in top 3 among national participants for technical product ideation and execution.
- **1st Rank, College Project Competition** — Awarded top position for engineering depth, real-world applicability, and presentation quality.